

f_4 : Dynkin Diagram



$$w(\mathbf{ad}) := [0001]$$

$$w(\mathbf{F}) := [1000]$$

Cartan Matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -2 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

Weight Metric

$$\begin{pmatrix} 1 & \frac{3}{2} & 2 & 1 \\ \frac{3}{2} & 3 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 2 \end{pmatrix}$$